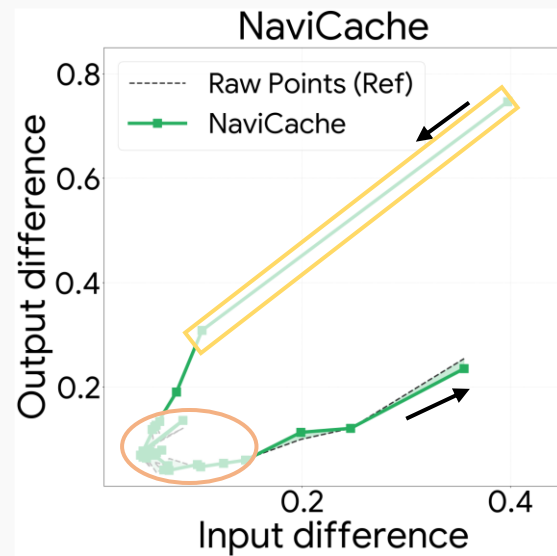
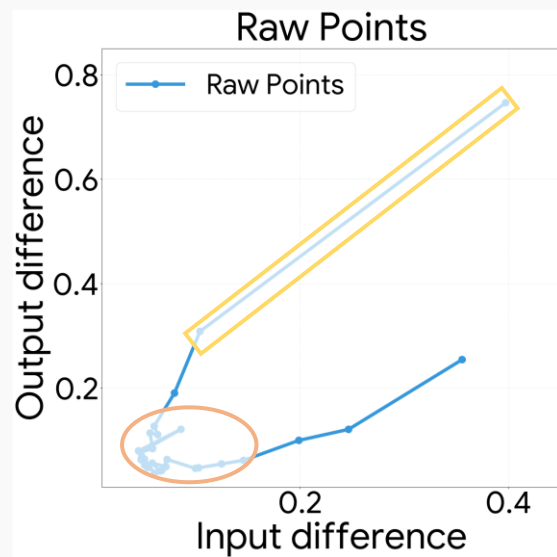




NaviCache



Initial Alignment

Full Computation for N_{align} steps
and
Initialize the NaviCache system

DiT Block
(Computed)

Measure
 $\Delta I, \Delta O$

Initialize
 $\hat{x}_0, P_0$

Test-Time Self-Calibration



Prior Projection

$$\hat{x}_{t|t-1} = \hat{x}_{t-1|t-1}$$

$$P_{t|t-1} = P_{t-1|t-1} + Q_t$$

Observational Rectification

$$K_t = P_{t|t-1}(P_{t|t-1} + R_t)^{-1}$$

$$\hat{x}_{t|t} = (I - K_t)\hat{x}_{t|t-1} + K_t z_t$$

$$P_{t|t} = (I - K_t)P_{t|t-1}$$

$$\hat{x}_{t|t} \leftarrow z_t = \Delta O_t / \Delta I_t$$

Uncertainty Estimation

Observation Error Prediction

$$\Delta \hat{O}_t = \hat{x}_{t|t-1} \Delta I_t$$

Accumulate Observation Error

$$\varepsilon_{acc} \leftarrow \varepsilon_{acc} + \Delta \hat{O}_t$$

NO $\varepsilon_{acc} < \tau?$ YES

UPDATE

SKIP

VDM Backbone

Prompt

Text
Embedding

VAE
Encoder

Initial
Noise

Initial Alignment
Stage

Self-Calibration
Stage

VAE
Decoder

Video

